Qn. Set Code-1

Semester: 4th
Programme: B.Tech
Branch: CSSE

SPRING END SEMESTER EXAMINATION-2023

4th Semester B.Tech

PROBABILITY AND STATISTICS

MA 2011

(For 2022 (L.E), 2021 & Previous Admitted Batches)

Time: 3 Hours Full Marks: 50

Answer any SIX questions.
Question paper consists of four SECTIONS i.e. A, B, C and D.
Section A is compulsory.
Attempt minimum one question each from Sections B, C, D.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) Let $P(A) = 0.7, P(B') = 0.5, P(A' \cup B') = 0.6$.
Find $P(A \cup B)$ where $A' = Complement\ of\ A$

(b) Find the value of k such that

$$f(x) = \begin{cases} kx(1-x), & 0 \le x \le 1 \\ 0, & \text{Otherwise} \end{cases}$$

is a probability density function.

(c) Let X be a random variable which follows the normal distribution with mean 0.8 and variance 4 then evaluate $P(1 \le X \le 1.8)$

(d) If X and Y are independent random variables with variances 2 and 3 respectively, then find $V(3X - 2Y)$.

(e) Let the random variable X has exponential distribution with variance 1, then find the 50th percentile of X.

(f) If $Y = aX + b,\ a \neq 0$ then find the $\text{Corr}(X, Y)$.

(g) Explain confidence interval and confidence level.

(h)

	y			
x	$p(x_i, y_j)$	0	5	10	15
	0	0.02	0.06	0.02	0.11
	5	0.04	0.13	0.19	0.10
	10	0.01	0.15	0.14	0.12

Find $P_Y(5), P_Y(10), P_Y(15)$.

(i) Find the standard deviation for Poisson distribution if the expected value of related random variable is 0.25.

(j) When circuit boards used in the manufacturer of compact disc players are tested, the long-run percentage of defective is 5%. Let $X =$ the number of defective boards in a random sample of size $n = 25$, i.e., $X \sim Bin(25, 0.05)$ where $Bin(n, p)$ is the binomial distribution with number of trials n and probability of success p. Determine $P(2 \leq X \leq 6)$.

SECTION-B

2. (a) A computer consulting firm presently has bids out on three projects. Let A_i = {awarded project i}, for i =1,2,3, and suppose that $P(A_1) = 0.28$, $P(A_2) = 0.23$, $P(A_3) = 0.32$, $P(A_1 \cap A_2) = 0.16$, $P(A_1 \cap A_3) = 0.08$, $P(A_2 \cap A_3) = 0.09$, $P(A_1 \cap A_2 \cap A_3) = 0.03$. Compute the probability of the following events: [4]

a) $A'_1 \cap A'_2$ b) $A'_1 \cap A'_2 \cap A'_3$

c) $A'_1 \cap A'_2 \cap A_3$ d) $(A'_1 \cap A'_2) \cup A_3$.

(b) Find mean and variance of geometric distribution. [4]

3. (a) Define correlation coefficient of two random variables X and Y and prove that [4]

$$Corr(aX + b, cY + d) = Corr(X, Y)$$

when a and c have the same sign. What happens if a and c have opposite sign?

(b) For any two events A and B show that [4]
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ by using Venn Diagram.

SECTION-C

4. (a) Find the maximum likelihood estimate of mean and variance of Poisson distribution. [4]

   (b) Find a 95% confidence interval for μ of a normal population with standard deviation 4 from the samples 66,66,65,64,66,67,64,65,63,64 and then find the length of the interval. [4]

5. (a) Define covariance of two random variables X and Y and show that $Cov(X,Y) = E(X,Y) - E(X)E(Y)$. [4]

   (b) A consumer organization that evaluates new automobile customarily reports the number of major defects in each car examined. Let X denote the number of major defects in a randomly selected car of a certain type. The cdf of X is as follows: [4]

$$F(x) = \begin{cases} 0 & x < 1 \\ 0.06 & 0 \le x < 1 \\ 0.19 & 1 \le x < 2 \\ 0.39 & 2 \le x < 3 \\ 0.67 & 3 \le x < 4 \\ 0.92 & 4 \le x < 5 \\ 0.95 & 5 \le x < 6. \\ 1 & 6 \le x \end{cases}$$

   a. Calculate the pmf $p(x)$ of X.

   b. Calculate the following probabilities directly from cdf, $P(2 < X < 5)$.

   c. Calculate the variance of X.

6. (a) The actual tracking weight of a stereo cartridge that is set to track at 3 g on a particular changer can be regarded as a continuous rv X with pdf [4]

$$f(x) = \begin{cases} k[1 - (x-3)^2] & 2 \le x \le 4 \\ 0 & \text{otherwise} \end{cases}$$

   a. Find the value of k.

b. Find the cumulative distribution function $F(x)$.

c. What is the probability that the actual tracking weight is within 0.25 g of the prescribed weight?

(b) A fair coin is tossed three times. Let X denotes the number of heads on the first two toss and Y denotes the number of heads on the last two toss. Find the $Corr(X,Y)$. [4]

SECTION-D

7. (a) An instructor has given a short quiz consisting of two parts. For a randomly selected student, let X =the number of points earned on the first part and Y =the number of points earned on the second part. Suppose that the joint pmf of X and Y is given in the accompanying table. [4]

		y			
p(x, y)		0	5	10	15
x	0	0.02	0.06	0.02	0.10
	5	0.04	0.15	0.20	0.10
	10	0.01	0.15	0.14	0.01

Compute the correlation coefficient ρ for this X and Y.

(b) The joint density function of (X,Y) is [4]

$$f(x,y) = \begin{cases} k(x+y^2), & 0 < x < 4, 0 < y < 2, \\ 0, & \text{otherwise} \end{cases}$$

Determine k and median of Y.

8. (a) a. Find the maximum likelihood estimate of θ if the density $f(x) = \begin{cases} \theta e^{-\theta x} & if \quad x \geq 0 \\ 0 & if \quad x < 0 \end{cases}$ [4]

(b) For the standardized normal distribution function $\Phi(z)$, prove that $\Phi(-z) = 1 - \Phi(z)$. [4]

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